

Base de datos: capacitor X2 según el color y la cantidad de LED

Red de **127 V / 60 Hz** · LED de 5 mm y 3 mm · fuente capacitiva de 5 canales (un capacitor por salida).

Cómo usarla

1. Busca el color y cuántos LED van **en esa salida**.
2. Elige el brillo: **bajo ≈ 10 mA** (los LED duran más), **medio ≈ 15 mA** o **alto ≈ 20 mA**, que es el brillo de la hoja de datos.
3. Si quieres exactamente 20 mA, usa la columna de **2 capacitores en paralelo**: los dos van en la misma bornera.
4. Con la luz alta (+10 %), el nivel alto nunca pasa de 24 mA. El LED aguanta 30 mA continuos.

Ejemplo

3 LED rojos (2.0 V típico, 2.3 V a alto brillo) en alto brillo: capacitor **474** = 0.47 μF → 19 mA. Para 20 mA exactos: **104 + 394** en paralelo.

El código del capacitor

474 = $47 \times 10^4 \text{ pF} = 0.47 \mu\text{F}$, no 47 faradios. **394** = 0.39 μF · **564** = 0.56 μF ·

684 = 0.68 μF · **105** = 1 μF .

La letra que sigue (J $\pm 5\%$, K $\pm 10\%$) es la tolerancia. Las dos sirven. "275 V~" es el voltaje, no el valor.

Fórmula

$I \approx 4 \cdot f \cdot C \cdot (V_{\text{pico}} - V_{\text{LED}})$, con $V_{\text{pico}} \approx 178 \text{ V}$. Ya incluye el LED testigo y la resistencia de 100 Ω de la versión 3. En las versiones 1 y 2 da unos 0.5 mA más; viene en su propia columna.

Siempre capacitor X2 de 275 VAC. Nunca cerámico ni de poliéster común.

Las celdas "—" no son posibles en una salida (demasiados LED). No sirve para LED de 1 W ni para RGB automáticos.

Hoja de datos por color (LED de 5 mm, valores típicos)

Color	Longitud de onda	Vf mín (V)	Vf típico (V)	Vf máx (V)	Corriente normal	Corriente máx. continua	Brillo típico (mcd, transparente)	Nota
Rojo	620–630	1.8	2	2.3	20 mA	30 mA	1 500–8 000	alto brillo a 20 mA ≈ 2.0–2.3 V
Naranja	600–610	1.8	2	2.3	20 mA	30 mA	2 000–8 000	
Amarillo	585–595	1.8	2.1	2.4	20 mA	30 mA	2 000–8 000	
Verde limón (GaP)	565–575	2	2.2	2.6	20 mA	30 mA	100–1 000	verde claro, poco brillo
Verde esmeralda	515–530	2.9	3.1	3.4	20 mA	30 mA	8 000–20 000	verde intenso (InGaN)
Azul	460–470	2.9	3.1	3.4	20 mA	30 mA	3 000–8 000	
Blanco frío	6000–7000 K	2.9	3.1	3.4	20 mA	30 mA	10 000–25 000	
Blanco cálido	2700–3500 K	2.9	3.1	3.4	20 mA	30 mA	8 000–18 000	
Rosa	chip azul + fósforo	2.9	3.1	3.4	20 mA	30 mA	3 000–8 000	
Violeta / UV	395–405	3	3.3	3.7	20 mA	30 mA	500–2 000	no mirar de frente

Valores típicos de LED comerciales; cambian un poco según la marca. La base de datos usa el Vf típico para la corriente y el Vf máximo para elegir el zener. Un LED de 3 mm tiene los mismos voltajes y la misma corriente.

Consulta rápida · brillo ALTO (≈20 mA)

LED por salida	Rojo	Naranja	Amarillo	Verde limón (GaP)	Verde esmeralda	Azul	Blanco frío	Blanco cálido	Rosa	Violeta / UV
1	474 19.4 mA	474 19.4 mA	474 19.4 mA	474 19.4 mA	474 19.3 mA	474 19.3 mA	474 19.3 mA	474 19.3 mA	474 19.3 mA	474 19.3 mA
2	474 19.2 mA	474 19.2 mA	474 19.2 mA	474 19.2 mA	474 19 mA	474 19 mA	474 19 mA	474 19 mA	474 19 mA	474 18.9 mA
3	474 19 mA	474 19 mA	474 19 mA	474 18.9 mA	474 18.6 mA	474 18.6 mA	474 18.6 mA	474 18.6 mA	474 18.6 mA	474 18.6 mA
4	474 18.8 mA	474 18.8 mA	474 18.7 mA	474 18.7 mA	474 18.3 mA	474 18.3 mA	474 18.3 mA	474 18.3 mA	474 18.3 mA	474 18.2 mA
5	474 18.5 mA	474 18.5 mA	474 18.5 mA	474 18.4 mA	564 21.4 mA	564 21.4 mA	564 21.4 mA	564 21.4 mA	564 21.4 mA	564 21.2 mA
6	474 18.3 mA	474 18.3 mA	474 18.3 mA	474 18.2 mA	564 20.9 mA	564 20.9 mA	564 20.9 mA	564 20.9 mA	564 20.9 mA	564 20.8 mA
7	564 21.6 mA	564 21.6 mA	564 21.5 mA	564 21.4 mA	564 20.5 mA	564 20.5 mA	564 20.5 mA	564 20.5 mA	564 20.5 mA	564 20.3 mA
8	564 21.3 mA	564 21.3 mA	564 21.2 mA	564 21.1 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 19.9 mA
9	564 21 mA	564 21 mA	564 20.9 mA	564 20.8 mA	564 19.7 mA	564 19.7 mA	564 19.7 mA	564 19.7 mA	564 19.7 mA	564 19.4 mA
10	564 20.8 mA	564 20.8 mA	564 20.6 mA	564 20.5 mA	564 19.3 mA	564 19.3 mA	564 19.3 mA	564 19.3 mA	564 19.3 mA	564 19 mA
12	564 20.2 mA	564 20.2 mA	564 20.1 mA	564 19.9 mA	564 18.4 mA	564 18.4 mA	564 18.4 mA	564 18.4 mA	564 18.4 mA	564 18.1 mA
15	564 19.4 mA	564 19.4 mA	564 19.2 mA	564 19 mA	684 20.9 mA	684 20.9 mA	684 20.9 mA	684 20.9 mA	684 20.9 mA	684 20.4 mA
18	564 18.6 mA	564 18.6 mA	564 18.4 mA	564 18.1 mA	684 19.4 mA	684 19.4 mA	684 19.4 mA	684 19.4 mA	684 19.4 mA	684 18.8 mA
20	564 18.1 mA	564 18.1 mA	564 17.8 mA	564 17.5 mA	684 18.3 mA	684 18.3 mA	684 18.3 mA	684 18.3 mA	684 18.3 mA	684 17.7 mA
25	684 20.3 mA	684 20.3 mA	684 19.9 mA	684 19.5 mA	824 19.1 mA	824 19.1 mA	824 19.1 mA	824 19.1 mA	824 19.1 mA	824 18.1 mA
30	684 18.7 mA	684 18.7 mA	684 18.2 mA	684 17.7 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	105 18.1 mA
35	684 17 mA	684 17 mA	824 19.9 mA	824 19.2 mA	105 15.8 mA	105 15.8 mA	105 15.8 mA	105 15.8 mA	105 15.8 mA	—
40	824 18.6 mA	824 18.6 mA	824 17.8 mA	824 17 mA	—	—	—	—	—	—

Consulta rápida · brillo MEDIO (≈15 mA)

LED por salida	Rojo	Naranja	Amarillo	Verde limón (GaP)	Verde esmeralda	Azul	Blanco frío	Blanco cálido	Rosa	Violeta / UV
1	394 16.1 mA	394 16.1 mA	394 16.1 mA	394 16.1 mA	394 16 mA	394 16 mA	394 16 mA	394 16 mA	394 16 mA	394 16 mA
2	394 15.9 mA	394 15.9 mA	394 15.9 mA	394 15.9 mA	394 15.7 mA	394 15.7 mA	394 15.7 mA	394 15.7 mA	394 15.7 mA	394 15.7 mA
3	394 15.8 mA	394 15.8 mA	394 15.7 mA	394 15.7 mA	394 15.5 mA	394 15.5 mA	394 15.5 mA	394 15.5 mA	394 15.5 mA	394 15.4 mA
4	394 15.6 mA	394 15.6 mA	394 15.5 mA	394 15.5 mA	394 15.2 mA	394 15.2 mA	394 15.2 mA	394 15.2 mA	394 15.2 mA	394 15.1 mA
5	394 15.4 mA	394 15.4 mA	394 15.3 mA	394 15.3 mA	394 14.9 mA	394 14.9 mA	394 14.9 mA	394 14.9 mA	394 14.9 mA	394 14.8 mA
6	394 15.2 mA	394 15.2 mA	394 15.1 mA	394 15.1 mA	394 14.6 mA	394 14.6 mA	394 14.6 mA	394 14.6 mA	394 14.6 mA	394 14.5 mA
7	394 15 mA	394 15 mA	394 14.9 mA	394 14.9 mA	394 14.3 mA	394 14.3 mA	394 14.3 mA	394 14.3 mA	394 14.3 mA	394 14.2 mA
8	394 14.8 mA	394 14.8 mA	394 14.8 mA	394 14.7 mA	394 14 mA	394 14 mA	394 14 mA	394 14 mA	394 14 mA	394 13.9 mA
9	394 14.6 mA	394 14.6 mA	394 14.6 mA	394 14.5 mA	394 13.7 mA	394 13.7 mA	394 13.7 mA	394 13.7 mA	394 13.7 mA	394 13.5 mA
10	394 14.5 mA	394 14.5 mA	394 14.4 mA	394 14.3 mA	394 13.4 mA	394 13.4 mA	394 13.4 mA	394 13.4 mA	394 13.4 mA	474 16 mA
12	394 14.1 mA	394 14.1 mA	394 14 mA	394 13.9 mA	474 15.5 mA	474 15.5 mA	474 15.5 mA	474 15.5 mA	474 15.5 mA	474 15.2 mA
15	394 13.5 mA	394 13.5 mA	394 13.4 mA	474 16 mA	474 14.4 mA	474 14.4 mA	474 14.4 mA	474 14.4 mA	474 14.4 mA	474 14.1 mA
18	474 15.6 mA	474 15.6 mA	474 15.4 mA	474 15.2 mA	474 13.4 mA	474 13.4 mA	474 13.4 mA	474 13.4 mA	474 13.4 mA	564 15.5 mA
20	474 15.2 mA	474 15.2 mA	474 14.9 mA	474 14.7 mA	564 15.1 mA	564 15.1 mA	564 15.1 mA	564 15.1 mA	564 15.1 mA	564 14.6 mA
25	474 14 mA	474 14 mA	474 13.8 mA	474 13.5 mA	564 13 mA	564 13 mA	564 13 mA	564 13 mA	564 13 mA	684 15 mA
30	564 15.4 mA	564 15.4 mA	564 15 mA	564 14.6 mA	684 13.3 mA	684 13.3 mA	684 13.3 mA	684 13.3 mA	684 13.3 mA	684 12.3 mA
35	564 14 mA	564 14 mA	564 13.6 mA	564 13.1 mA	824 13 mA	824 13 mA	824 13 mA	824 13 mA	824 13 mA	—
40	564 12.7 mA	564 12.7 mA	684 14.8 mA	684 14.1 mA	—	—	—	—	—	—

Consulta rápida · brillo BAJO (≈10 mA, larga vida)

LED por salida	Rojo	Naranja	Amarillo	Verde limón (GaP)	Verde esmeralda	Azul	Blanco frío	Blanco cálido	Rosa	Violeta / UV
1	224 9.1 mA	224 9.1 mA	224 9.1 mA	224 9.1 mA	224 9 mA	224 9 mA	224 9 mA	224 9 mA	224 9 mA	224 9 mA
2	224 9 mA	224 9 mA	224 9 mA	224 9 mA	224 8.9 mA	224 8.9 mA	224 8.9 mA	224 8.9 mA	224 8.9 mA	224 8.9 mA
3	224 8.9 mA	224 8.9 mA	224 8.9 mA	224 8.9 mA	274 10.7 mA	274 10.7 mA	274 10.7 mA	274 10.7 mA	274 10.7 mA	274 10.7 mA
4	274 10.8 mA	274 10.8 mA	274 10.8 mA	274 10.7 mA	274 10.5 mA	274 10.5 mA	274 10.5 mA	274 10.5 mA	274 10.5 mA	274 10.4 mA
5	274 10.7 mA	274 10.7 mA	274 10.6 mA	274 10.6 mA	274 10.3 mA	274 10.3 mA	274 10.3 mA	274 10.3 mA	274 10.3 mA	274 10.2 mA
6	274 10.5 mA	274 10.5 mA	274 10.5 mA	274 10.4 mA	274 10.1 mA	274 10.1 mA	274 10.1 mA	274 10.1 mA	274 10.1 mA	274 10 mA
7	274 10.4 mA	274 10.4 mA	274 10.3 mA	274 10.3 mA	274 9.9 mA	274 9.9 mA	274 9.9 mA	274 9.9 mA	274 9.9 mA	274 9.8 mA
8	274 10.3 mA	274 10.3 mA	274 10.2 mA	274 10.2 mA	274 9.7 mA	274 9.7 mA	274 9.7 mA	274 9.7 mA	274 9.7 mA	274 9.6 mA
9	274 10.1 mA	274 10.1 mA	274 10.1 mA	274 10 mA	274 9.5 mA	274 9.5 mA	274 9.5 mA	274 9.5 mA	274 9.5 mA	274 9.4 mA
10	274 10 mA	274 10 mA	274 9.9 mA	274 9.9 mA	274 9.3 mA	274 9.3 mA	274 9.3 mA	274 9.3 mA	274 9.3 mA	274 9.2 mA
12	274 9.7 mA	274 9.7 mA	274 9.7 mA	274 9.6 mA	274 8.9 mA	274 8.9 mA	274 8.9 mA	274 8.9 mA	274 8.9 mA	274 8.7 mA
15	274 9.4 mA	274 9.4 mA	274 9.3 mA	274 9.2 mA	334 10.1 mA	334 10.1 mA	334 10.1 mA	334 10.1 mA	334 10.1 mA	334 9.9 mA
18	274 9 mA	274 9 mA	274 8.9 mA	274 8.7 mA	334 9.4 mA	334 9.4 mA	334 9.4 mA	334 9.4 mA	334 9.4 mA	334 9.1 mA
20	274 8.7 mA	274 8.7 mA	334 10.5 mA	334 10.3 mA	334 8.9 mA	334 8.9 mA	334 8.9 mA	334 8.9 mA	334 8.9 mA	394 10.1 mA
25	334 9.9 mA	334 9.9 mA	334 9.7 mA	334 9.5 mA	394 9.1 mA	394 9.1 mA	394 9.1 mA	394 9.1 mA	394 9.1 mA	394 8.6 mA
30	334 9.1 mA	334 9.1 mA	334 8.8 mA	394 10.1 mA	474 9.2 mA	474 9.2 mA	474 9.2 mA	474 9.2 mA	474 9.2 mA	474 8.5 mA
35	394 9.8 mA	394 9.8 mA	394 9.4 mA	394 9.1 mA	564 8.9 mA	564 8.9 mA	564 8.9 mA	564 8.9 mA	564 8.9 mA	—
40	394 8.8 mA	394 8.8 mA	394 8.5 mA	474 9.7 mA	—	—	—	—	—	—

Consulta rápida · 20 mA exactos con 2 capacitores en paralelo

LED por salida	Rojo	Naranja	Amarillo	Verde limón (GaP)	Verde esmeralda	Azul	Blanco frío	Blanco cálido	Rosa	Violeta / UV
1	154 + 334 19.9 mA	154 + 334 19.9 mA	154 + 334 19.8 mA	154 + 334 19.8 mA	104 + 394 20.1 mA	104 + 394 20.1 mA	104 + 394 20.1 mA	104 + 394 20.1 mA	104 + 394 20.1 mA	104 + 394 20.1 mA
2	104 + 394 20 mA	104 + 394 20 mA	104 + 394 20 mA	104 + 394 20 mA	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.7 mA
3	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.8 mA	104 + 394 19.7 mA	124 + 394 20.2 mA	124 + 394 20.2 mA	124 + 394 20.2 mA	124 + 394 20.2 mA	124 + 394 20.2 mA	124 + 394 20.1 mA
4	124 + 394 20.4 mA	124 + 394 20.4 mA	124 + 394 20.3 mA	124 + 394 20.3 mA	124 + 394 19.8 mA	124 + 394 19.8 mA	124 + 394 19.8 mA	124 + 394 19.8 mA	124 + 394 19.8 mA	124 + 394 19.7 mA
5	124 + 394 20.1 mA	124 + 394 20.1 mA	124 + 394 20.1 mA	124 + 394 20 mA	124 + 394 19.4 mA	124 + 394 19.4 mA	124 + 394 19.4 mA	124 + 394 19.4 mA	124 + 394 19.4 mA	154 + 394 20.5 mA
6	124 + 394 19.9 mA	124 + 394 19.9 mA	124 + 394 19.8 mA	124 + 394 19.7 mA	154 + 394 20.2 mA	154 + 394 20.2 mA	154 + 394 20.2 mA	154 + 394 20.2 mA	154 + 394 20.2 mA	154 + 394 20 mA
7	124 + 394 19.6 mA	124 + 394 19.6 mA	124 + 394 19.5 mA	124 + 394 19.5 mA	224 + 334 20.2 mA	224 + 334 20.2 mA	224 + 334 20.2 mA	224 + 334 20.2 mA	224 + 334 20.2 mA	224 + 334 20 mA
8	154 + 394 20.5 mA	154 + 394 20.5 mA	154 + 394 20.4 mA	154 + 394 20.3 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 20.1 mA	564 19.9 mA
9	154 + 394 20.3 mA	154 + 394 20.3 mA	154 + 394 20.2 mA	154 + 394 20 mA	104 + 474 20 mA	104 + 474 20 mA	104 + 474 20 mA	104 + 474 20 mA	104 + 474 20 mA	184 + 394 19.8 mA
10	154 + 394 20 mA	154 + 394 20 mA	154 + 394 19.9 mA	224 + 334 20.1 mA	124 + 474 20.3 mA	124 + 474 20.3 mA	124 + 474 20.3 mA	124 + 474 20.3 mA	124 + 474 20.3 mA	124 + 474 20 mA
12	224 + 334 19.9 mA	224 + 334 19.9 mA	564 20.1 mA	564 19.9 mA	224 + 394 20.1 mA	224 + 394 20.1 mA	224 + 394 20.1 mA	224 + 394 20.1 mA	224 + 394 20.1 mA	154 + 474 20.1 mA
15	184 + 394 19.8 mA	184 + 394 19.8 mA	124 + 474 20.2 mA	124 + 474 20 mA	184 + 474 20 mA	184 + 474 20 mA	184 + 474 20 mA	184 + 474 20 mA	184 + 474 20 mA	104 + 564 19.8 mA
18	274 + 334 19.9 mA	274 + 334 19.9 mA	224 + 394 20 mA	154 + 474 20.1 mA	154 + 564 20.2 mA	154 + 564 20.2 mA	154 + 564 20.2 mA	154 + 564 20.2 mA	154 + 564 20.2 mA	334 + 394 19.9 mA
20	154 + 474 20 mA	154 + 474 20 mA	154 + 474 19.7 mA	184 + 474 20.3 mA	184 + 564 20 mA	184 + 564 20 mA	184 + 564 20 mA	184 + 564 20 mA	184 + 564 20 mA	104 + 684 20.3 mA
25	104 + 564 19.7 mA	104 + 564 19.7 mA	684 19.9 mA	224 + 474 19.8 mA	394 + 474 20 mA	394 + 474 20 mA	394 + 474 20 mA	394 + 474 20 mA	394 + 474 20 mA	224 + 684 19.9 mA
30	334 + 394 19.8 mA	334 + 394 19.8 mA	184 + 564 19.8 mA	104 + 684 20.3 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	105 19.5 mA	394 + 684 19.4 mA
35	124 + 684 20 mA	124 + 684 20 mA	154 + 684 20.1 mA	394 + 474 20.1 mA	184 + 105 18.7 mA	184 + 105 18.7 mA	184 + 105 18.7 mA	184 + 105 18.7 mA	184 + 105 18.7 mA	—
40	184 + 684 19.5 mA	184 + 684 19.5 mA	104 + 824 20 mA	274 + 684 19.7 mA	—	—	—	—	—	—

Tabla inversa · corriente (mA) de cada capacitor según el voltaje que suman los LED

Suma los volts de los LED de la salida (por ejemplo, 5 rojos × 2.0 = 10 V) y busca la columna. En verde: 18–24 mA (alto brillo). En rojo: más de 24 mA, demasiado para LED de 5 mm.

Código X2	µF	5 V	10 V	15 V	20 V	30 V	40 V	50 V	60 V	75 V	90 V	105 V	120 V
104	0.1	4.2	4	3.9	3.8	3.6	3.3	3.1	2.8	2.5	2.1	1.8	1.4
124	0.12	5	4.8	4.7	4.6	4.3	4	3.7	3.4	3	2.5	2.1	1.7
154	0.15	6.2	6.1	5.9	5.7	5.3	5	4.6	4.3	3.7	3.2	2.6	2.1
184	0.18	7.5	7.3	7.1	6.8	6.4	6	5.5	5.1	4.5	3.8	3.2	2.5
224	0.22	9.1	8.9	8.6	8.4	7.8	7.3	6.8	6.2	5.4	4.7	3.9	3.1
274	0.27	11.2	10.9	10.6	10.3	9.6	9	8.3	7.7	6.7	5.7	4.7	3.8
334	0.33	13.7	13.3	12.9	12.5	11.7	10.9	10.2	9.4	8.2	7	5.8	4.6
394	0.39	16.2	15.7	15.3	14.8	13.9	12.9	12	11.1	9.7	8.3	6.9	5.4
474	0.47	19.5	19	18.4	17.8	16.7	15.6	14.5	13.3	11.6	9.9	8.3	6.6
564	0.56	23.3	22.6	21.9	21.3	19.9	18.6	17.2	15.9	13.9	11.9	9.8	7.8
684	0.68	28.3	27.5	26.6	25.8	24.2	22.6	20.9	19.3	16.8	14.4	11.9	9.5
824	0.82	34.1	33.1	32.1	31.1	29.2	27.2	25.2	23.3	20.3	17.4	14.4	11.5
105	1	41.6	40.4	39.2	38	35.6	33.2	30.8	28.4	24.8	21.2	17.6	14
125	1.2	49.9	48.4	47	45.6	42.7	39.8	36.9	34	29.7	25.4	21.1	16.8
155	1.5	62.4	60.6	58.8	57	53.4	49.8	46.2	42.6	37.2	31.8	26.4	21

Base de datos completa (173 combinaciones)

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
1	Rojo	2	1	2	224	9.1	394	16.1	474	19.4	21.5	19.9	154 + 334	19.9	1N5349B (12 V)	0.04
2	Rojo	2	2	4	224	9	394	15.9	474	19.2	21.2	19.7	104 + 394	20	1N5349B (12 V)	0.08
3	Rojo	2	3	6	224	8.9	394	15.8	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.11
4	Rojo	2	4	8	274	10.8	394	15.6	474	18.8	20.8	19.2	124 + 394	20.4	1N5355B (18 V)	0.15
5	Rojo	2	5	10	274	10.7	394	15.4	474	18.5	20.6	19	124 + 394	20.1	1N5357B (20 V)	0.18
6	Rojo	2	6	12	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.9	1N5358B (22 V)	0.22
7	Rojo	2	7	14	274	10.4	394	15	564	21.6	24	22.1	124 + 394	19.6	1N5359B (24 V)	0.3
8	Rojo	2	8	16	274	10.3	394	14.8	564	21.3	23.7	21.8	154 + 394	20.5	1N5361B (27 V)	0.34
9	Rojo	2	9	18	274	10.1	394	14.6	564	21	23.4	21.5	154 + 394	20.3	1N5363B (30 V)	0.38
10	Rojo	2	10	20	274	10	394	14.5	564	20.8	23.2	21.3	154 + 394	20	1N5365B (36 V)	0.42
11	Rojo	2	12	24	274	9.7	394	14.1	564	20.2	22.6	20.7	224 + 334	19.9	1N5366B (39 V)	0.48
12	Rojo	2	15	30	274	9.4	394	13.5	564	19.4	21.8	19.9	184 + 394	19.8	1N5368B (47 V)	0.58
13	Rojo	2	18	36	274	9	474	15.6	564	18.6	21	19.1	274 + 334	19.9	1N5370B (56 V)	0.67
14	Rojo	2	20	40	274	8.7	474	15.2	564	18.1	20.5	18.6	154 + 474	20	1N5372B (62 V)	0.72
15	Rojo	2	25	50	334	9.9	474	14	684	20.3	23.2	20.9	104 + 564	19.7	1N5374B (75 V)	1.01
16	Rojo	2	30	60	334	9.1	564	15.4	684	18.7	21.6	19.3	334 + 394	19.8	1N5377B (91 V)	1.12
17	Rojo	2	35	70	394	9.8	564	14	684	17	20	17.7	124 + 684	20	1N5378B (100 V)	1.19
18	Rojo	2	40	80	394	8.8	564	12.7	824	18.6	22.1	19.3	184 + 684	19.5	1N5380B (120 V)	1.49
19	Naranja	2	1	2	224	9.1	394	16.1	474	19.4	21.5	19.9	154 + 334	19.9	1N5349B (12 V)	0.04
20	Naranja	2	2	4	224	9	394	15.9	474	19.2	21.2	19.7	104 + 394	20	1N5349B (12 V)	0.08
21	Naranja	2	3	6	224	8.9	394	15.8	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.11
22	Naranja	2	4	8	274	10.8	394	15.6	474	18.8	20.8	19.2	124 + 394	20.4	1N5355B (18 V)	0.15
23	Naranja	2	5	10	274	10.7	394	15.4	474	18.5	20.6	19	124 + 394	20.1	1N5357B (20 V)	0.18
24	Naranja	2	6	12	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.9	1N5358B (22 V)	0.22
25	Naranja	2	7	14	274	10.4	394	15	564	21.6	24	22.1	124 + 394	19.6	1N5359B (24 V)	0.3
26	Naranja	2	8	16	274	10.3	394	14.8	564	21.3	23.7	21.8	154 + 394	20.5	1N5361B (27 V)	0.34
27	Naranja	2	9	18	274	10.1	394	14.6	564	21	23.4	21.5	154 + 394	20.3	1N5363B (30 V)	0.38
28	Naranja	2	10	20	274	10	394	14.5	564	20.8	23.2	21.3	154 + 394	20	1N5365B (36 V)	0.42
29	Naranja	2	12	24	274	9.7	394	14.1	564	20.2	22.6	20.7	224 + 334	19.9	1N5366B (39 V)	0.48
30	Naranja	2	15	30	274	9.4	394	13.5	564	19.4	21.8	19.9	184 + 394	19.8	1N5368B (47 V)	0.58
31	Naranja	2	18	36	274	9	474	15.6	564	18.6	21	19.1	274 + 334	19.9	1N5370B (56 V)	0.67
32	Naranja	2	20	40	274	8.7	474	15.2	564	18.1	20.5	18.6	154 + 474	20	1N5372B (62 V)	0.72
33	Naranja	2	25	50	334	9.9	474	14	684	20.3	23.2	20.9	104 + 564	19.7	1N5374B (75 V)	1.01

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
34	Naranja	2	30	60	334	9.1	564	15.4	684	18.7	21.6	19.3	334 + 394	19.8	1N5377B (91 V)	1.12
35	Naranja	2	35	70	394	9.8	564	14	684	17	20	17.7	124 + 684	20	1N5378B (100 V)	1.19
36	Naranja	2	40	80	394	8.8	564	12.7	824	18.6	22.1	19.3	184 + 684	19.5	1N5380B (120 V)	1.49
37	Amarillo	2.1	1	2.1	224	9.1	394	16.1	474	19.4	21.5	19.9	154 + 334	19.8	1N5349B (12 V)	0.04
38	Amarillo	2.1	2	4.2	224	9	394	15.9	474	19.2	21.2	19.6	104 + 394	20	1N5349B (12 V)	0.08
39	Amarillo	2.1	3	6.3	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12
40	Amarillo	2.1	4	8.4	274	10.8	394	15.5	474	18.7	20.8	19.2	124 + 394	20.3	1N5355B (18 V)	0.16
41	Amarillo	2.1	5	10.5	274	10.6	394	15.3	474	18.5	20.5	18.9	124 + 394	20.1	1N5357B (20 V)	0.19
42	Amarillo	2.1	6	12.6	274	10.5	394	15.1	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
43	Amarillo	2.1	7	14.7	274	10.3	394	14.9	564	21.5	23.9	22	124 + 394	19.5	1N5361B (27 V)	0.32
44	Amarillo	2.1	8	16.8	274	10.2	394	14.8	564	21.2	23.6	21.7	154 + 394	20.4	1N5361B (27 V)	0.36
45	Amarillo	2.1	9	18.9	274	10.1	394	14.6	564	20.9	23.3	21.4	154 + 394	20.2	1N5363B (30 V)	0.4
46	Amarillo	2.1	10	21	274	9.9	394	14.4	564	20.6	23	21.1	154 + 394	19.9	1N5365B (36 V)	0.43
47	Amarillo	2.1	12	25.2	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.51
48	Amarillo	2.1	15	31.5	274	9.3	394	13.4	564	19.2	21.6	19.7	124 + 474	20.2	1N5368B (47 V)	0.6
49	Amarillo	2.1	18	37.8	274	8.9	474	15.4	564	18.4	20.8	18.9	224 + 394	20	1N5370B (56 V)	0.7
50	Amarillo	2.1	20	42	334	10.5	474	14.9	564	17.8	20.2	18.3	154 + 474	19.7	1N5372B (62 V)	0.75
51	Amarillo	2.1	25	52.5	334	9.7	474	13.8	684	19.9	22.8	20.5	684	19.9	1N5374B (75 V)	1.04
52	Amarillo	2.1	30	63	334	8.8	564	15	684	18.2	21.1	18.8	184 + 564	19.8	1N5377B (91 V)	1.15
53	Amarillo	2.1	35	73.5	394	9.4	564	13.6	824	19.9	23.4	20.6	154 + 684	20.1	1N5379B (110 V)	1.46
54	Amarillo	2.1	40	84	394	8.5	684	14.8	824	17.8	21.3	18.5	104 + 824	20	1N5380B (120 V)	1.5
55	Verde limón (GaP)	2.2	1	2.2	224	9.1	394	16.1	474	19.4	21.5	19.9	154 + 334	19.8	1N5349B (12 V)	0.04
56	Verde limón (GaP)	2.2	2	4.4	224	9	394	15.9	474	19.2	21.2	19.6	104 + 394	20	1N5349B (12 V)	0.08
57	Verde limón (GaP)	2.2	3	6.6	224	8.9	394	15.7	474	18.9	21	19.4	104 + 394	19.7	1N5352B (15 V)	0.12
58	Verde limón (GaP)	2.2	4	8.8	274	10.7	394	15.5	474	18.7	20.7	19.1	124 + 394	20.3	1N5355B (18 V)	0.16
59	Verde limón (GaP)	2.2	5	11	274	10.6	394	15.3	474	18.4	20.5	18.9	124 + 394	20	1N5357B (20 V)	0.2
60	Verde limón (GaP)	2.2	6	13.2	274	10.4	394	15.1	474	18.2	20.2	18.6	124 + 394	19.7	1N5359B (24 V)	0.24
61	Verde limón (GaP)	2.2	7	15.4	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.5	1N5361B (27 V)	0.33
62	Verde limón (GaP)	2.2	8	17.6	274	10.2	394	14.7	564	21.1	23.5	21.6	154 + 394	20.3	1N5363B (30 V)	0.37

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
63	Verde limón (GaP)	2.2	9	19.8	274	10	394	14.5	564	20.8	23.2	21.3	154 + 394	20	1N5365B (36 V)	0.41
64	Verde limón (GaP)	2.2	10	22	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.1	1N5365B (36 V)	0.45
65	Verde limón (GaP)	2.2	12	26.4	274	9.6	394	13.9	564	19.9	22.3	20.4	564	19.9	1N5367B (43 V)	0.53
66	Verde limón (GaP)	2.2	15	33	274	9.2	474	16	564	19	21.4	19.5	124 + 474	20	1N5369B (51 V)	0.63
67	Verde limón (GaP)	2.2	18	39.6	274	8.7	474	15.2	564	18.1	20.5	18.6	154 + 474	20.1	1N5372B (62 V)	0.72
68	Verde limón (GaP)	2.2	20	44	334	10.3	474	14.7	564	17.5	19.9	18	184 + 474	20.3	1N5373B (68 V)	0.77
69	Verde limón (GaP)	2.2	25	55	334	9.5	474	13.5	684	19.5	22.4	20.1	224 + 474	19.8	1N5375B (82 V)	1.07
70	Verde limón (GaP)	2.2	30	66	394	10.1	564	14.6	684	17.7	20.6	18.3	104 + 684	20.3	1N5378B (100 V)	1.17
71	Verde limón (GaP)	2.2	35	77	394	9.1	564	13.1	824	19.2	22.7	19.9	394 + 474	20.1	1N5379B (110 V)	1.48
72	Verde limón (GaP)	2.2	40	88	474	9.7	684	14.1	824	17	20.5	17.8	274 + 684	19.7	1N5381B (130 V)	1.5
73	Verde esmeralda	3.1	1	3.1	224	9	394	16	474	19.3	21.3	19.8	104 + 394	20.1	1N5349B (12 V)	0.06
74	Verde esmeralda	3.1	2	6.2	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12
75	Verde esmeralda	3.1	3	9.3	274	10.7	394	15.5	474	18.6	20.6	19.1	124 + 394	20.2	1N5355B (18 V)	0.17
76	Verde esmeralda	3.1	4	12.4	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
77	Verde esmeralda	3.1	5	15.5	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.4	1N5361B (27 V)	0.33
78	Verde esmeralda	3.1	6	18.6	274	10.1	394	14.6	564	20.9	23.4	21.5	154 + 394	20.2	1N5363B (30 V)	0.39
79	Verde esmeralda	3.1	7	21.7	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.2	1N5365B (36 V)	0.44
80	Verde esmeralda	3.1	8	24.8	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.5
81	Verde esmeralda	3.1	9	27.9	274	9.5	394	13.7	564	19.7	22.1	20.2	104 + 474	20	1N5367B (43 V)	0.55
82	Verde esmeralda	3.1	10	31	274	9.3	394	13.4	564	19.3	21.7	19.8	124 + 474	20.3	1N5368B (47 V)	0.6
83	Verde esmeralda	3.1	12	37.2	274	8.9	474	15.5	564	18.4	20.9	19	224 + 394	20.1	1N5370B (56 V)	0.68
84	Verde esmeralda	3.1	15	46.5	334	10.1	474	14.4	684	20.9	23.8	21.5	184 + 474	20	1N5373B (68 V)	0.97
85	Verde esmeralda	3.1	18	55.8	334	9.4	474	13.4	684	19.4	22.3	20	154 + 564	20.2	1N5375B (82 V)	1.08
86	Verde esmeralda	3.1	20	62	334	8.9	564	15.1	684	18.3	21.3	19	184 + 564	20	1N5377B (91 V)	1.13
87	Verde esmeralda	3.1	25	77.5	394	9.1	564	13	824	19.1	22.6	19.8	394 + 474	20	1N5379B (110 V)	1.48
88	Verde esmeralda	3.1	30	93	474	9.2	684	13.3	105	19.5	23.8	20.4	105	19.5	1N5381B (130 V)	1.81
89	Verde esmeralda	3.1	35	108.5	564	8.9	824	13	105	15.8	20.1	16.7	184 + 105	18.7	1N5383B (150 V)	1.71
90	Azul	3.1	1	3.1	224	9	394	16	474	19.3	21.3	19.8	104 + 394	20.1	1N5349B (12 V)	0.06
91	Azul	3.1	2	6.2	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
92	Azul	3.1	3	9.3	274	10.7	394	15.5	474	18.6	20.6	19.1	124 + 394	20.2	1N5355B (18 V)	0.17
93	Azul	3.1	4	12.4	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
94	Azul	3.1	5	15.5	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.4	1N5361B (27 V)	0.33
95	Azul	3.1	6	18.6	274	10.1	394	14.6	564	20.9	23.4	21.5	154 + 394	20.2	1N5363B (30 V)	0.39
96	Azul	3.1	7	21.7	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.2	1N5365B (36 V)	0.44
97	Azul	3.1	8	24.8	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.5
98	Azul	3.1	9	27.9	274	9.5	394	13.7	564	19.7	22.1	20.2	104 + 474	20	1N5367B (43 V)	0.55
99	Azul	3.1	10	31	274	9.3	394	13.4	564	19.3	21.7	19.8	124 + 474	20.3	1N5368B (47 V)	0.6
100	Azul	3.1	12	37.2	274	8.9	474	15.5	564	18.4	20.9	19	224 + 394	20.1	1N5370B (56 V)	0.68
101	Azul	3.1	15	46.5	334	10.1	474	14.4	684	20.9	23.8	21.5	184 + 474	20	1N5373B (68 V)	0.97
102	Azul	3.1	18	55.8	334	9.4	474	13.4	684	19.4	22.3	20	154 + 564	20.2	1N5375B (82 V)	1.08
103	Azul	3.1	20	62	334	8.9	564	15.1	684	18.3	21.3	19	184 + 564	20	1N5377B (91 V)	1.13
104	Azul	3.1	25	77.5	394	9.1	564	13	824	19.1	22.6	19.8	394 + 474	20	1N5379B (110 V)	1.48
105	Azul	3.1	30	93	474	9.2	684	13.3	105	19.5	23.8	20.4	105	19.5	1N5381B (130 V)	1.81
106	Azul	3.1	35	108.5	564	8.9	824	13	105	15.8	20.1	16.7	184 + 105	18.7	1N5383B (150 V)	1.71
107	Blanco frío	3.1	1	3.1	224	9	394	16	474	19.3	21.3	19.8	104 + 394	20.1	1N5349B (12 V)	0.06
108	Blanco frío	3.1	2	6.2	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12
109	Blanco frío	3.1	3	9.3	274	10.7	394	15.5	474	18.6	20.6	19.1	124 + 394	20.2	1N5355B (18 V)	0.17
110	Blanco frío	3.1	4	12.4	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
111	Blanco frío	3.1	5	15.5	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.4	1N5361B (27 V)	0.33
112	Blanco frío	3.1	6	18.6	274	10.1	394	14.6	564	20.9	23.4	21.5	154 + 394	20.2	1N5363B (30 V)	0.39
113	Blanco frío	3.1	7	21.7	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.2	1N5365B (36 V)	0.44
114	Blanco frío	3.1	8	24.8	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.5
115	Blanco frío	3.1	9	27.9	274	9.5	394	13.7	564	19.7	22.1	20.2	104 + 474	20	1N5367B (43 V)	0.55
116	Blanco frío	3.1	10	31	274	9.3	394	13.4	564	19.3	21.7	19.8	124 + 474	20.3	1N5368B (47 V)	0.6
117	Blanco frío	3.1	12	37.2	274	8.9	474	15.5	564	18.4	20.9	19	224 + 394	20.1	1N5370B (56 V)	0.68
118	Blanco frío	3.1	15	46.5	334	10.1	474	14.4	684	20.9	23.8	21.5	184 + 474	20	1N5373B (68 V)	0.97
119	Blanco frío	3.1	18	55.8	334	9.4	474	13.4	684	19.4	22.3	20	154 + 564	20.2	1N5375B (82 V)	1.08
120	Blanco frío	3.1	20	62	334	8.9	564	15.1	684	18.3	21.3	19	184 + 564	20	1N5377B (91 V)	1.13
121	Blanco frío	3.1	25	77.5	394	9.1	564	13	824	19.1	22.6	19.8	394 + 474	20	1N5379B (110 V)	1.48
122	Blanco frío	3.1	30	93	474	9.2	684	13.3	105	19.5	23.8	20.4	105	19.5	1N5381B (130 V)	1.81
123	Blanco frío	3.1	35	108.5	564	8.9	824	13	105	15.8	20.1	16.7	184 + 105	18.7	1N5383B (150 V)	1.71
124	Blanco cálido	3.1	1	3.1	224	9	394	16	474	19.3	21.3	19.8	104 + 394	20.1	1N5349B (12 V)	0.06

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
125	Blanco cálido	3.1	2	6.2	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12
126	Blanco cálido	3.1	3	9.3	274	10.7	394	15.5	474	18.6	20.6	19.1	124 + 394	20.2	1N5355B (18 V)	0.17
127	Blanco cálido	3.1	4	12.4	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
128	Blanco cálido	3.1	5	15.5	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.4	1N5361B (27 V)	0.33
129	Blanco cálido	3.1	6	18.6	274	10.1	394	14.6	564	20.9	23.4	21.5	154 + 394	20.2	1N5363B (30 V)	0.39
130	Blanco cálido	3.1	7	21.7	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.2	1N5365B (36 V)	0.44
131	Blanco cálido	3.1	8	24.8	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.5
132	Blanco cálido	3.1	9	27.9	274	9.5	394	13.7	564	19.7	22.1	20.2	104 + 474	20	1N5367B (43 V)	0.55
133	Blanco cálido	3.1	10	31	274	9.3	394	13.4	564	19.3	21.7	19.8	124 + 474	20.3	1N5368B (47 V)	0.6
134	Blanco cálido	3.1	12	37.2	274	8.9	474	15.5	564	18.4	20.9	19	224 + 394	20.1	1N5370B (56 V)	0.68
135	Blanco cálido	3.1	15	46.5	334	10.1	474	14.4	684	20.9	23.8	21.5	184 + 474	20	1N5373B (68 V)	0.97
136	Blanco cálido	3.1	18	55.8	334	9.4	474	13.4	684	19.4	22.3	20	154 + 564	20.2	1N5375B (82 V)	1.08
137	Blanco cálido	3.1	20	62	334	8.9	564	15.1	684	18.3	21.3	19	184 + 564	20	1N5377B (91 V)	1.13
138	Blanco cálido	3.1	25	77.5	394	9.1	564	13	824	19.1	22.6	19.8	394 + 474	20	1N5379B (110 V)	1.48
139	Blanco cálido	3.1	30	93	474	9.2	684	13.3	105	19.5	23.8	20.4	105	19.5	1N5381B (130 V)	1.81
140	Blanco cálido	3.1	35	108.5	564	8.9	824	13	105	15.8	20.1	16.7	184 + 105	18.7	1N5383B (150 V)	1.71
141	Rosa	3.1	1	3.1	224	9	394	16	474	19.3	21.3	19.8	104 + 394	20.1	1N5349B (12 V)	0.06
142	Rosa	3.1	2	6.2	224	8.9	394	15.7	474	19	21	19.4	104 + 394	19.8	1N5352B (15 V)	0.12
143	Rosa	3.1	3	9.3	274	10.7	394	15.5	474	18.6	20.6	19.1	124 + 394	20.2	1N5355B (18 V)	0.17
144	Rosa	3.1	4	12.4	274	10.5	394	15.2	474	18.3	20.3	18.7	124 + 394	19.8	1N5358B (22 V)	0.23
145	Rosa	3.1	5	15.5	274	10.3	394	14.9	564	21.4	23.8	21.9	124 + 394	19.4	1N5361B (27 V)	0.33
146	Rosa	3.1	6	18.6	274	10.1	394	14.6	564	20.9	23.4	21.5	154 + 394	20.2	1N5363B (30 V)	0.39
147	Rosa	3.1	7	21.7	274	9.9	394	14.3	564	20.5	22.9	21	224 + 334	20.2	1N5365B (36 V)	0.44
148	Rosa	3.1	8	24.8	274	9.7	394	14	564	20.1	22.5	20.6	564	20.1	1N5366B (39 V)	0.5
149	Rosa	3.1	9	27.9	274	9.5	394	13.7	564	19.7	22.1	20.2	104 + 474	20	1N5367B (43 V)	0.55
150	Rosa	3.1	10	31	274	9.3	394	13.4	564	19.3	21.7	19.8	124 + 474	20.3	1N5368B (47 V)	0.6
151	Rosa	3.1	12	37.2	274	8.9	474	15.5	564	18.4	20.9	19	224 + 394	20.1	1N5370B (56 V)	0.68
152	Rosa	3.1	15	46.5	334	10.1	474	14.4	684	20.9	23.8	21.5	184 + 474	20	1N5373B (68 V)	0.97
153	Rosa	3.1	18	55.8	334	9.4	474	13.4	684	19.4	22.3	20	154 + 564	20.2	1N5375B (82 V)	1.08
154	Rosa	3.1	20	62	334	8.9	564	15.1	684	18.3	21.3	19	184 + 564	20	1N5377B (91 V)	1.13
155	Rosa	3.1	25	77.5	394	9.1	564	13	824	19.1	22.6	19.8	394 + 474	20	1N5379B (110 V)	1.48
156	Rosa	3.1	30	93	474	9.2	684	13.3	105	19.5	23.8	20.4	105	19.5	1N5381B (130 V)	1.81
157	Rosa	3.1	35	108.5	564	8.9	824	13	105	15.8	20.1	16.7	184 + 105	18.7	1N5383B (150 V)	1.71

ID	Color	Vf	LED	V total	Bajo ≈10 mA	mA	Medio ≈15 mA	mA	Alto ≈20 mA	mA	mA red +10 %	mA en V1/V2	Exacto 20 mA (2 en paralelo)	mA	Zener 5 W (V3)	W LED
158	Violeta / UV	3.3	1	3.3	224	9	394	16	474	19.3	21.3	19.7	104 + 394	20.1	1N5349B (12 V)	0.06
159	Violeta / UV	3.3	2	6.6	224	8.9	394	15.7	474	18.9	21	19.4	104 + 394	19.7	1N5352B (15 V)	0.12
160	Violeta / UV	3.3	3	9.9	274	10.7	394	15.4	474	18.6	20.6	19	124 + 394	20.1	1N5355B (18 V)	0.18
161	Violeta / UV	3.3	4	13.2	274	10.4	394	15.1	474	18.2	20.2	18.6	124 + 394	19.7	1N5358B (22 V)	0.24
162	Violeta / UV	3.3	5	16.5	274	10.2	394	14.8	564	21.2	23.6	21.7	154 + 394	20.5	1N5361B (27 V)	0.35
163	Violeta / UV	3.3	6	19.8	274	10	394	14.5	564	20.8	23.2	21.3	154 + 394	20	1N5365B (36 V)	0.41
164	Violeta / UV	3.3	7	23.1	274	9.8	394	14.2	564	20.3	22.7	20.8	224 + 334	20	1N5365B (36 V)	0.47
165	Violeta / UV	3.3	8	26.4	274	9.6	394	13.9	564	19.9	22.3	20.4	564	19.9	1N5366B (39 V)	0.53
166	Violeta / UV	3.3	9	29.7	274	9.4	394	13.5	564	19.4	21.9	20	184 + 394	19.8	1N5368B (47 V)	0.58
167	Violeta / UV	3.3	10	33	274	9.2	474	16	564	19	21.4	19.5	124 + 474	20	1N5369B (51 V)	0.63
168	Violeta / UV	3.3	12	39.6	274	8.7	474	15.2	564	18.1	20.5	18.6	154 + 474	20.1	1N5370B (56 V)	0.72
169	Violeta / UV	3.3	15	49.5	334	9.9	474	14.1	684	20.4	23.3	21	104 + 564	19.8	1N5374B (75 V)	1.01
170	Violeta / UV	3.3	18	59.4	334	9.1	564	15.5	684	18.8	21.7	19.4	334 + 394	19.9	1N5375B (82 V)	1.12
171	Violeta / UV	3.3	20	66	394	10.1	564	14.6	684	17.7	20.6	18.3	104 + 684	20.3	1N5377B (91 V)	1.17
172	Violeta / UV	3.3	25	82.5	394	8.6	684	15	824	18.1	21.6	18.8	224 + 684	19.9	1N5380B (120 V)	1.49
173	Violeta / UV	3.3	30	99	474	8.5	684	12.3	105	18.1	22.4	19	394 + 684	19.4	1N5383B (150 V)	1.79

Archivo para Excel: **base_datos_capacitores_led.csv**.